

Assignment 3 - Part 3.1: 2D Manifold Latent Space Visualization & Analysis

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1 Introduction

This experiment explores how an Autoencoder compresses high-dimensional data into a low-dimensional latent space. We train a symmetric MNIST Autoencoder with a 2D bottleneck ($N = 2$) independently, then visualize the latent representations of the test set to analyze how the network organizes different digit classes.

2 Methodology

2.1 Architecture

A symmetric undercomplete Autoencoder:

- **Encoder:** $784 \rightarrow 128$ (ReLU) $\rightarrow 64$ (ReLU) $\rightarrow 2$ (Bottleneck)
- **Decoder:** $2 \rightarrow 64$ (ReLU) $\rightarrow 128$ (ReLU) $\rightarrow 784$ (Sigmoid)

2.2 Training Details

- **Dataset:** MNIST (60,000 train / 10,000 test), pixels in $[0, 1]$
- **Optimizer:** Adam ($lr = 10^{-3}$)
- **Loss:** MSE Reconstruction Loss
- **Epochs:** 30
- **Batch Size:** 256

3 Results

3.1 Reconstruction Loss

Figure 1 shows the MSE loss decreasing steadily over 30 epochs, confirming convergence.

3.2 Original vs Reconstructed Images

Figure 2 shows sample test images and their reconstructions. With only 2 latent dimensions, reconstructions are blurry but preserve the overall digit shape.

3.3 2D Latent Space Scatter Plot

Figure 3 visualizes the 2D latent codes of the entire MNIST test set, colored by true digit label.

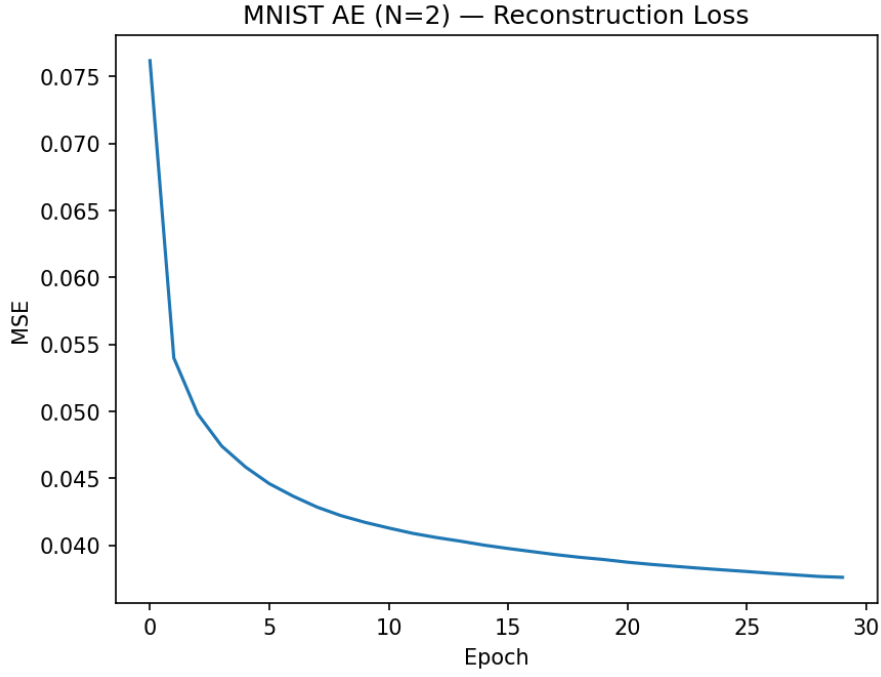


Figure 1: Reconstruction Loss (MSE) over 30 epochs for the MNIST AE with $N = 2$.

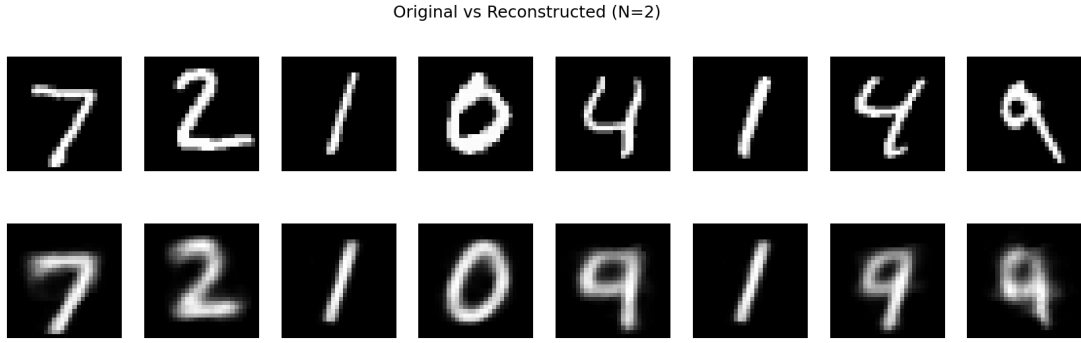


Figure 2: Original (top) vs Reconstructed (bottom) MNIST digits from the $N = 2$ Autoencoder.

4 Analysis

1. **Clustering:** Digits naturally form clusters in the 2D latent space, even though the AE was trained without any label information. This shows the network learns a meaningful representation purely from reconstruction.
2. **Similar Digits Overlap:** Visually similar digits (e.g., 4 & 9, 3 & 8, 1 & 7) tend to cluster nearby or partially overlap. This is because they share structural features (e.g., similar stroke patterns), so the encoder maps them to nearby latent coordinates.
3. **Distinct Digits Separate:** Structurally distinct digits (e.g., 0 and 1) are well-separated, confirming the latent space captures meaningful visual differences.

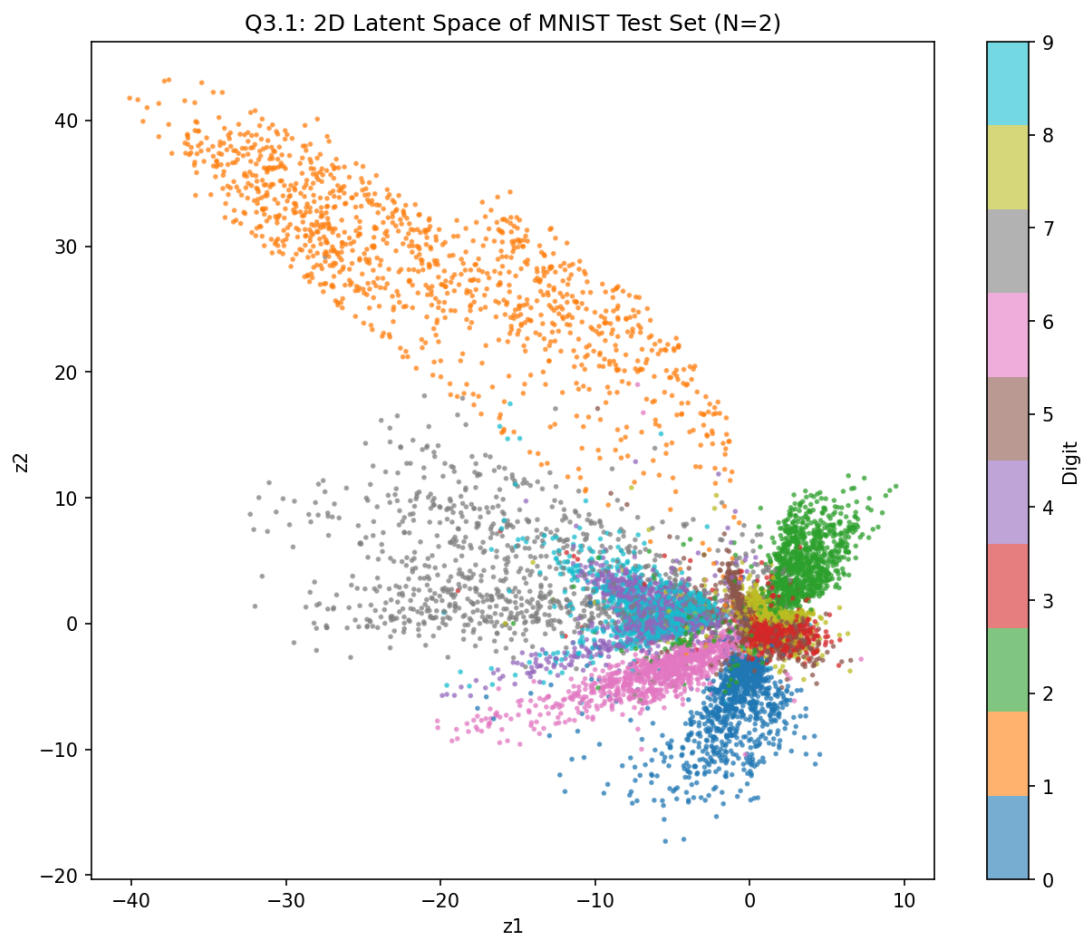


Figure 3: 2D Latent Space of MNIST test set. Each point is an image encoded to 2D, colored by its digit label (0–9).